new. Even Einstein complained about this, and how it helped keep the Copenhagen interpretation entrenched. “This state of affairs will last for many more years,” he wrote in 1951, “mainly because physicists have no understanding of logical and philosophical arguments.” And in most ways, education and access to education are far better now than they ever have been before. But with the massive increase in knowledge and information in the last century, education became unavoidably specialized. There’s nothing wrong with that, but with specialization comes boundaries on knowledge, and good specialists understand that. Indeed, it’s unlikely that Hawking, Tyson, or Krauss would issue strong pronouncements in many areas where they have no background—say, parasite ecology or best practices in industrial sheet metal production. So why do they feel so comfortable saying so much about philosophy? Why is philosophy held in such contempt by many physicists (and other scientists of all stripes)?

Philosophy has an image problem. Philosophers are thought to be mystics, religious figures, bullshit artists—anything divorced from reality. The discipline as a whole is seen as millennia of people chasing down big questions—What is the meaning of life? Why is there suffering?—and coming back without any good answers. Philosophers of physics, and most other philosophers, are far removed from this picture: they work on well-defined questions with logical rigor and with input from the most recent developments in science and from the immediate experiences of the senses. How the practice and the image of philosophy have diverged so wildly is a subject for an entirely different book, but one part of the answer probably lies in the split between the two major branches of modern Western philosophy, *analytic* and *Continental* philosophy. (Those names are largely historical accidents and not relevant to their work.) How the two branches split is a long and complex story (it has to do with the debate between the positivists and the German idealists, which we touched on in Chapter 8), but while most philosophers of physics are analytic, most of the philosophers from the past seventy years that you’ve heard of are probably Continental. Continental philosophers like Sartre, Camus, Foucault, Derrida, and Žižek have become public figures, while very few analytic philosophers have done the same. And Continental philosophers tend to be much more suspicious of scientific claims about knowledge and truth than are their analytic colleagues. Yet the distinction between the two kinds of